.....[2]

**6 (a)** A dust particle is suspended in the air between two parallel plates with the direction of electric field shown in Fig. 6.1. The electric field strength between the two parallel plate is $3.0 \times 10^{3}$ $Vm^{-1}$. The particle has charges of $+1.4 \times 10^{-15}$ C and $-1.4 \times 10^{-15}$ C near its ends. The charges may be considered to be static point charges separated by a distance of 3.0 mm, as shown in Fig. 6.1.

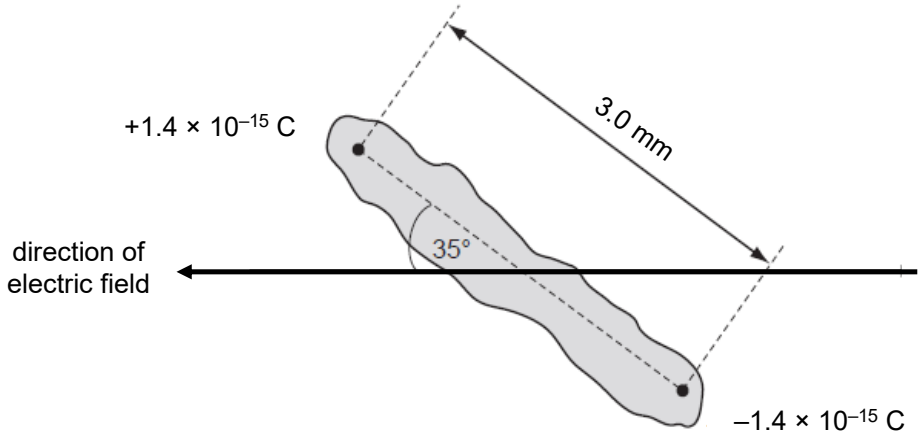


**Fig. 6.1**

**[Turn over**

The particle makes an angle of 35° with the direction of the electric field.

**(i)** Calculate the magnitude of the force on each charge due to the electric field.

force =…….. N [1]

**(ii)** Describe and explain the subsequent motion of the particle in the electric field.

……

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…… [3]

**6 (b)** Fig. 6.2 and Fig 6.3 show the electric field pattern near two point charges respectively.

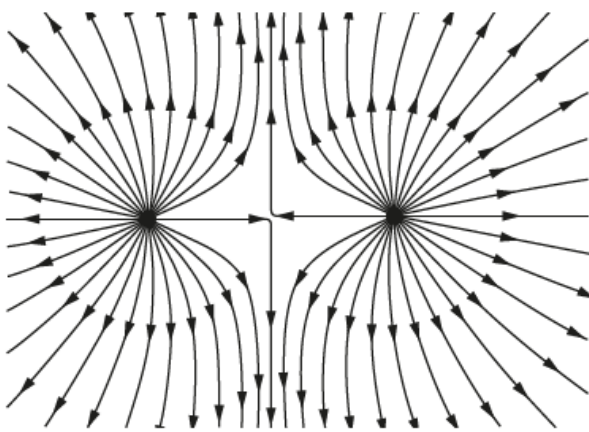

**Fig. 6.2**

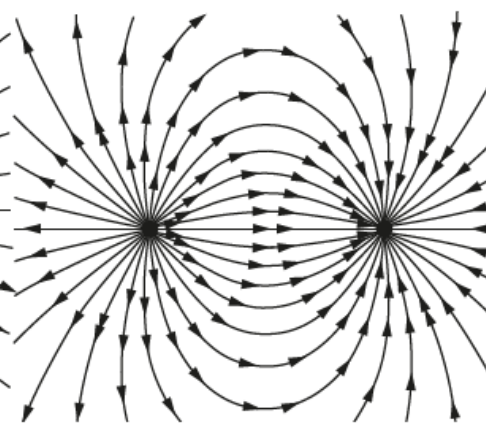

**Fig. 6.3**

**(i)** On the appropriate figure(s) mark the point(s)

**1.** with zero electric field and label it with the letter "N"

**2.** with zero electric potential and label it with the letter "E" [1]

**(ii)** On Fig. 6.2, draw **three** complete equipotential lines.

[2]

**6** **(c)** Fig. 6.4 shows the electrical potential $V$ and the magnitude of the electric field $E$ against distance $R$ for an isolated – 1.0 C charge.

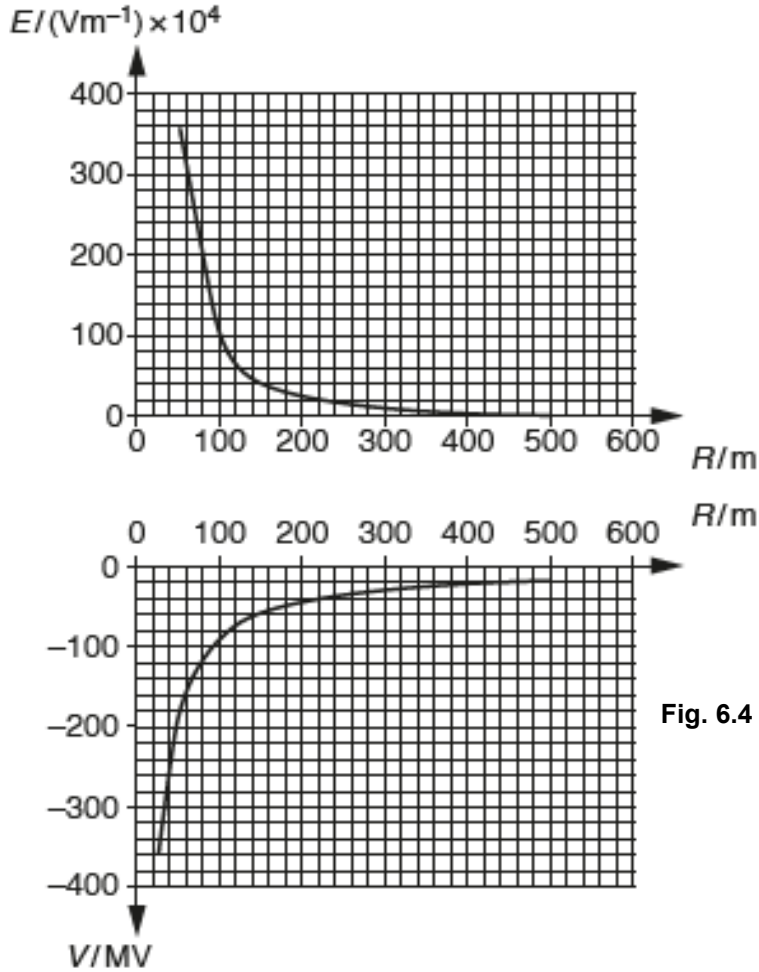


**Fig. 6.4**

Use Fig. 6.4 to explain the relationship between the electric field strength and the electric potential. (You may annotate on the figure and make clear reference to these annotations to support your answer.)

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……[4]

**[Turn over**